# ScanCrypt

Post-Quantum Cryptographic Security Assessment

| | |
|---|---|
| **Project Name** | ref |
| **Analysis Date** | 2026-07-09 20:52:22 |
| **ScanCrypt Version** | 1.0 |
| **Algorithm** | ML-DSA / Dilithium |
| **Security Level** | ML-DSA-44 / Dilithium2 |
| **Implementation Type** | Masked Implementation |

**Medium**

## Confidentiality Notice

This document contains confidential assessment details based on ScanCrypt analysis. Unauthorized disclosure is prohibited.

Report generated automatically by ScanCrypt.

# Executive Summary

| | | | |
|---|---|---|---|
| **Overall Risk** | Medium | **Algorithm** | ML-DSA / Dilithium |
| **Security Level** | ML-DSA-44 / Dilithium2 | **Implementation Type** | Masked Implementation |
| **Programming Language** | C/C++ | **Build System** | CMake |
| **Files** | 39 | **Source Files** | 20 |
| **Header Files** | 17 | **Lines of Code** | 5802 |
| **Rules Executed** | 0 | **Rules Passed** | 0 |
| **Rules Failed** | 0 | **Analysis Confidence** | 99% |
| **Detection Confidence** | 99% | **Analysis Duration** | N/A |

**Critical: 0**

**High: 0**

**Medium: 0**

**Low: 0**

**Information: 0**

# Project Inspection

| | |
|---|---|
| **Algorithm** | ML-DSA / Dilithium |
| **Security Level** | ML-DSA-44 / Dilithium2 |
| **Implementation Type** | Masked Implementation |
| **Programming Language** | C/C++ |
| **Build System** | CMake |
| **Files** | 39 |
| **Source Files** | 20 |
| **Header Files** | 17 |
| **Lines of Code** | 5802 |
| **Detection Confidence** | 99% |

## Detection Evidence

- NTT implementation detected in ntt.c
- montgomery_reduce detected in ntt.c
- DILITHIUM2 detected in api.h
- DILITHIUM3 detected in api.h
- DILITHIUM5 detected in api.h
- montgomery_reduce detected in reduce.h
- Polynomial arithmetic detected in poly.h
- make_hint detected in poly.h
- use_hint detected in poly.h
- decompose detected in poly.h
- poly detected in poly.h
- NTT implementation detected in poly.h
- make_hint detected in rounding.h
- use_hint detected in rounding.h
- decompose detected in rounding.h
- DILITHIUM_MODE detected in params.h
- montgomery_reduce detected in reduce.c
- Polynomial arithmetic detected in sign.c
- make_hint detected in sign.c
- use_hint detected in sign.c

- decompose detected in sign.c
- poly detected in sign.c
- pack_pk detected in sign.c
- pack_sk detected in sign.c
- NTT implementation detected in sign.c
- SHAKE / SHA-3 primitive detected in sign.c
- fips202 detected in sign.c
- SHAKE / SHA-3 primitive detected in symmetric-shake.c
- fips202 detected in symmetric-shake.c
- DILITHIUM_MODE detected in config.h
- DILITHIUM2 detected in config.h
- DILITHIUM3 detected in config.h
- DILITHIUM5 detected in config.h
- CRYPTO_ALGNAME detected in config.h
- Polynomial arithmetic detected in polyvec.c
- make_hint detected in polyvec.c
- use_hint detected in polyvec.c
- decompose detected in polyvec.c
- poly detected in polyvec.c
- NTT implementation detected in polyvec.c
- SHAKE / SHA-3 primitive detected in polyvec.c
- Polynomial arithmetic detected in packing.c
- poly detected in packing.c
- pack_pk detected in packing.c
- pack_sk detected in packing.c
- SHAKE / SHA-3 primitive detected in fips202.h
- fips202 detected in fips202.h
- DILITHIUM_MODE detected in Makefile
- DILITHIUM2 detected in Makefile
- DILITHIUM3 detected in Makefile
- DILITHIUM5 detected in Makefile
- poly detected in Makefile
- NTT implementation detected in Makefile
- fips202 detected in Makefile
- SHAKE / SHA-3 primitive detected in Makefile
- Polynomial arithmetic detected in test_vectors.c
- make_hint detected in test_vectors.c
- decompose detected in test_vectors.c
- poly detected in test_vectors.c

- NTT implementation detected in test_vectors.c
- SHAKE / SHA-3 primitive detected in test_vectors.c
- fips202 detected in test_vectors.c
- Polynomial arithmetic detected in packing.h
- pack_pk detected in packing.h
- pack_sk detected in packing.h
- poly detected in test_mul.c
- NTT implementation detected in test_mul.c
- SHAKE / SHA-3 primitive detected in fips202.c
- fips202 detected in fips202.c
- Polynomial arithmetic detected in polyvec.h
- make_hint detected in polyvec.h
- use_hint detected in polyvec.h
- decompose detected in polyvec.h
- poly detected in polyvec.h
- NTT implementation detected in polyvec.h
- make_hint detected in rounding.c
- use_hint detected in rounding.c
- decompose detected in rounding.c
- Polynomial arithmetic detected in poly.c
- make_hint detected in poly.c
- use_hint detected in poly.c
- decompose detected in poly.c
- rej_uniform detected in poly.c
- poly detected in poly.c
- NTT implementation detected in poly.c
- montgomery_reduce detected in poly.c
- SHAKE / SHA-3 primitive detected in poly.c
- NTT implementation detected in ntt.h
- poly detected in sign.h
- CRYPTO_ALGNAME detected in PQCgenKAT_sign.c
- Polynomial arithmetic detected in test_speed.c
- poly detected in test_speed.c
- NTT implementation detected in test_speed.c
- DILITHIUM_MODE detected in CMakeLists.txt
- DILITHIUM2 detected in CMakeLists.txt
- DILITHIUM3 detected in CMakeLists.txt
- DILITHIUM5 detected in CMakeLists.txt
- poly detected in CMakeLists.txt

- NTT implementation detected in CMakeLists.txt
- fips202 detected in CMakeLists.txt
- SHAKE / SHA-3 primitive detected in symmetric.h
- fips202 detected in symmetric.h
- K = 4
- L = 4
- ETA = 2
- Masked Implementation indicators detected in /tmp/scancrypt_upload_v0hz3i_8/ntt.c
- Masked Implementation indicators detected in /tmp/scancrypt_upload_v0hz3i_8/aes256ctr.c
- Masked Implementation indicators detected in /tmp/scancrypt_upload_v0hz3i_8/fips202.c

# Findings Summary

| Rule ID | Severity | Category | Status | File | Line |
|---|---|---|---|---|---|
| No findings | | | | | |

# Detailed Findings

No findings were detected during analysis.

# Rule Execution Summary

No rule execution data available.

# Statistics

**Total rules executed**
0

**Passed rules**
0

**Failed rules**
0

**Analysis duration**
N/A

**Files analysed**
39

**Source lines analysed**
5802

## Execution Progress

Passed: 0 / 1

Failed: 0 / 1

# Appendix

## Analysed Files

ref/ntt.c

ref/precomp.gp

ref/randombytes.c

ref/api.h

ref/reduce.h

ref/poly.h

ref/rounding.h

ref/params.h

ref/reduce.c

ref/sign.c

ref/symmetric-shake.c

ref/test/cpucycles.c

ref/test/speed_print.c

ref/test/test_vectors.c

ref/test/cpucycles.h

ref/test/test_mul.c

ref/test/.gitignore

ref/test/test_dilithium.c

ref/test/speed_print.h

ref/test/test_speed.c

ref/config.h

ref/polyvec.c

ref/packing.c

ref/aes256ctr.c

ref/fips202.h

ref/Makefile

ref/randombytes.h

ref/rng.h

ref/packing.h

ref/symmetric-aes.c

ref/aes256ctr.h

ref/.gitignore

ref/fips202.c

ref/polyvec.h

ref/rounding.c

ref/poly.c

ref/ntt.h

ref/sign.h

ref/PQCgenKAT_sign.c

ref/rng.c

ref/CMakeLists.txt

ref/symmetric.h

**Tool version**
1.0

**Report generated**
2026-07-09 20:52:22

**Scan configuration**
Default ScanCrypt profile

**Supported algorithms**
ML-DSA (Dilithium), ML-KEM (Kyber)

**Supported implementation types**
Portable C / Reference, AVX2 Optimized, ARM Cortex-M / Embedded, Masked Implementation, Hardware / FPGA